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FEDERAL SPECIFICATION

POZZOLAN (FOR USE IN PORTLAND-CEMENT CONCRETE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers pozzolan for use in portland-cement concrete.

1.2 Classification. Pozzolan shall be either class N or class F as specified (see 6.2 and 6.3).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

Federal Specifications:

SS-C-192 - Cement, Portland.
UU-S-48 - Sacks; Paper, Shipping.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 158 - Cements, Hydraulic; Sampling, Inspection, and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402).

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington).

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies).

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspections by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

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American Society for Testing and Materials, (ASTM) Standards:

C227 - Method of test for Potential Alkali Reactivity of Cement-Aggregate Combinations (Mortar Bar Method).

C311 - Method of Sampling and Testing Fly Ash for Use as an Admixture in Portland Cement Concrete.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

American Chemical Society

Reagent Chemicals, ACS Specifications.

(Application for copies should be addressed to the American Chemical Society, 1155 16th Street, N. W., Washington, D. C. 20036.)

Uniform Classification Committee:

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, Ill. 60606.)

American Trucking Associations Inc.

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc. 1616 P Street, N. W., Washington, D. C. 20036.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Materials. Pozzolan may be produced from any selection of raw materials and by any process that will yield a product conforming to the requirements of this specification.

3.2 Chemical and physical. Pozzolan shall conform to the applicable requirements in tables I and II.

TABLE I. Chemical requirements

Chemical requirement	Class	
	N	F
Silicon dioxide (SiO_2), plus aluminum oxide (Al_2O_3), plus iron oxide (Fe_2O_3), min., percent.	75.0	75.0
Magnesium oxide (MgO), max., percent.	5.0	5.0
Sulfur trioxide (SO_3), max., percent.	4.0	4.0
Loss on ignition, max., percent.	8.0	6.0
Moisture content, max., percent.	3.0	3.0
Available alkalies, max., percent.	2.0	2.0

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TABLE II. Physical requirements

Physical requirement	Class N	Class F
Fineness, surface area, sq cm/cu cm, min.	12,000	6,500
Pozzolanic strength of cement-pozzolan mortar at 28 days, min., percent of control	80	75
Compressive strength of lime-pozzolan mortar at 7 days, min. p.s.i.	900	900
Increase in drying shrinkage of mortar bars at 28 days, max., percent	0.04	---
Soundness, autoclave expansion, max., percent 1/	0.50	0.50
Water requirements, increase in flow (over control, at stipulated water content), min. 2/	0	0
Reactivity with cement alkalies, reduction of mortar expansion at 14 days min., percent. 3/	60	60

1/ The specimen shall remain firm and hard and show no signs of distortion, cracking, checking, pitting, or disintegration after having been subjected to the autoclave expansion test.

2/ This requires that a mortar with a class N pozzolan and 112 percent of the water content of the control mortar, or a mortar with a class F pozzolan and 103 percent of the water content of the control shall have a flow equal to or greater than that of the control.

3/ This requirement applies only when specified (see 6.2).

3.3 Uniformity. Pozzolan shall conform to the following uniformity requirements (see 4.4.3.1 and 4.4.3.8).

Fineness, surface area, maximum variation from average, percent = 25
Specific gravity, maximum variation from average, percent = 6

These requirements will apply not only to samples within a lot but also to samples in consecutive lots. They will apply to all samples taken regardless of whether the material in any given lot or lots is shipped to a single project or to more than one project.

Results of tests of fineness and specific gravity on any individual sample shall not vary be more than the specified amounts, expressed as percentages of the average, from the average of similar tests on the five immediately preceding samples that comply in all respects with the applicable provisions of these specifications.

In computing the average, the results for samples that do not meet all applicable requirements of these specifications shall be omitted regardless of the disposition that may have been made of the material represented by such samples.

The uniformity requirements will be invoked only after five samples have been tested and found to fully comply with the specifications. The uniformity requirements will apply to the sixth such sample tested, and to all subsequent samples; the results of these tests on the sixth such sample will be compared with the average of the results of these tests on all preceding samples; the results of these tests on the seventh such and all subsequent samples will be compared with the results of these tests on the five preceding samples found to fully comply with the specifications.

When, in the judgment of the specifying agency, a major interruption in the delivery of material from a source has occurred, or when there is a permitted change in the source, the application of the uniformity requirements will be terminated, and reinvoked after five samples have been tested and found to comply with all applicable provisions of these specifications.

3.4 Indentification marking of packages. When the pozzolan is delivered in packages, the word "Pozzolan for use in Portland-Cement Concrete," the name of the producer, and the weight of the pozzolan contained in the package shall be plainly marked on each package. Similar information shall be provided in the shipping documents accompanying the shipment of the packaged or bulk pozzolan.

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3.5 Storage. The pozzolan shall be stored in weathertight bins or silos at the source in such a manner as to permit easy access for proper inspection and indentification of such shipment. Every facility shall be provided for representatives of the Government to perform careful sampling and inspecification, as may be required (see 4.1), both at the source and at the site of the work.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Lot. Unless otherwise specified (see 6.2), a lot shall consist of all pozzolan of the same class offered for delivery at the same time.

4.2.2 Sampling for examination of packing. Packed pozzolan shall be sampled in accordance with level S-3 in accordance with MIL-STD-105.

4.2.3 Sampling for test. Samples of pozzolan taken for testing shall weigh at least 4 lb. (2 kg) each. One test sample shall represent not more than 100 tons.

4.2.4 Methods. The pozzolan may be sampled by any one of the following methods, as approved by the specifying agency for use at the particular storage facility employed.

4.2.4.1 From the conveyor delivering to bulk storage at the source. One sample of at least 4 lb. (2 kg) shall be taken from each 100 tons or less passing over the conveyor. This may be secured by taking the entire sample at a single operation, known as the "grab method", or by combining several portions taken at regular intervals, known as the "composite method". A composite sample shall be obtained by combining and thoroughly mixing approximately equal weights of the pozzolan taken at regular intervals, each portion representing approximately 2 tons, to provide test samples each weighing at least 4 lb. and each representing not more than 100 tons. Automatic samplers may be used in obtaining samples.

4.2.4.2 From source bulk storage at points of discharge. Sufficient pozzolan shall be drawn from the discharge openings to obtain samples representative of the pozzolan as determined by the appearance, at the discharge openings, of indicators placed on the surface of the pozzolan directly above these openings before the drawing of the pozzolan is started. One sample weighing at least 4 lb. (2 kg) shall be taken for each 100 tons or less by either the grab or composite method as discribed in 4.2.4.1 above. If the bulk storage container is equipped with side discharge openings so arranged that each is opposite a volume of the contents of the container that weighs not more than 100 tons, a sample weighing not less that 4 lb. taken from each such opening may be considered as complying with this requirement.

4.2.4.3 From bulk storage by proper sampling tubes. When the methods described in 4.2.4.1 and 4.2.4.2 cannot be applied, and when the depth of the pozzolan to be sampled does not exceed 10 ft. (3 m), samples may be obtained by the use of proper sampling tubes inserted vertically to the full depth of the pozzolan. Samples so taken shall be obtained from points well distributed over the storage area.

4.2.4.4 Other cases. In all other cases, samples shall be taken from each 2 tons or portions thereof in the lot and combined to form test samples weighing at least 4 lb. and representing not more than 100 tons. In case of samples from cars and trucks, when the pozzolan is being shipped from one source, it is permissible to combine the samples from several trucks to form a test sample representing not more than 100 tons.

4.3 Examination.

4.3.1 End item. Packed pozzolan shall be examined for the defects listed in table III. The acceptable quality levels (AQL's) for defects listed in table III shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects, in accordance with MIL-STD-105.

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TABLE III. Classification of defects

Defects	Major	Minor
Bags not of the grade specified.	X	
Bags not fully closed.	X	
Bags split or torn.	X	
Hole in bags.	X	
Marking ommitted or illegible.	X	
Marking incomplete.		X
Marking in wrong location.		X

4.3.2 Examination of preparation for delivery. The packing and marking of the pozzolan shall be examined to determine compliance with the requirements of Section 5.

4.4 Test methods.

4.4.1 Preparation of samples for test. Composite samples for tests, as required below, shall be prepared by arranging all test samples in groups, each group representing 100 tons. From each of the test samples in a group, equal portions shall be taken, sufficient in amount to form a composite sample large enough to permit making the required tests. The composite sample thus prepared shall be thoroughly mixed before using. A similar procedure, using samples which each represent 100 tons, shall be followed to prepare composite samples representing 1000 tons.

4.4.1.2 Number of tests. Each sample representing 100 tons, or the sample representing the quantity sampled when this is less than 100 tons, will be tested for:

Loss ignition	Water requirement
Moisture content	Specific gravity
Fineness, surface area	Strength of lime-pozzolan mortar

4.4.1.3 Other tests. All other physical tests and the remainder of the chemical analysis will be made on composite samples representing each 1000 tons, the composite sample for this purpose being prepared by combining equal parts of 10 consecutive samples, each representing 100 tons. When the amount sampled is less than 1000 tons, all tests will be made on a composite sample representing the quantity sampled. The composite samples should be thoroughly mixed before using.

4.4.2 Chemical tests. Tests for chemical composition shall be made in accordance with the following methods and conform to the applicable provisions of Method 1001 of Fed. Test Method Std. No. 158. Reagent-grade chemicals shall be used in all tests. Unless otherwise indicated, it is intended that all reagents used shall conform to the specifications of the Committee on Analytical Reagents of the American Chemical Society, where such specifications are available. Other grades may be used, provided it is first ascertained that the reagent is of sufficiently high purity to permit its use without lessening the accuracy of the determination. Unless otherwise indicated, references to water in the following paragraphs shall be understood to mean distilled water. Concentrations of reagents used in these methods shall conform to Section 3.2 of Method 1001 of Fed. Test Method Std. No. 158.

4.4.2.1 Moisture content. Dry a weighed sample of the pozzolan, in the condition received for test, in a porcelain crucible to constant weight in an oven at 105° to 110°C, cool, and weigh. Calculate the percentage of moisture to the nearest 0.1 percent as follows:

$$\text{Moisture content, percent} = \frac{A}{B} \times 100$$

where:

A = weight loss caused by drying, g, and
B = weight before drying, g.

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4.4.2.2 Loss on ignition. Determine loss on ignition in accordance with paragraph 2.1 or 2.2 Method 1301 of Fed. Test Method Std. No. 158, except that the material remaining from the determination of moisture content shall be ignited to constant weight in an uncovered porcelain (not platinum) crucible at 700° to 800°C for fly ash, and at 900° to 1000°C. for natural pozzolan. Calculate the loss on ignition to the nearest 0.1 percent as follows:

$$\text{Loss on ignition, percent} = \frac{A}{B} \times 100$$

where:

A = weight loss between 105°C. and ignition temperature, g, and
B = weight of moisture-free sample used, g.

4.4.2.3 Sum of silicon dioxide, aluminum oxide, and iron oxide.

4.4.2.3.1 Referee method. Weigh out a sample of ignited material equivalent to 0.5 g of moisture-free sample before ignition, calculated as follows:

$$\text{Weight of ignited sample, g} = \frac{0.5\ (100.0 - \text{loss on ignition, percent})}{100}$$

Thoroughly mix the sample with 4 to 6 g of Na_2CO_3 by grinding in an agate mortar. Place a thin layer of Na_2CO_3 on the bottom of 20- to 30-ml platinum crucible, add the mixture, and cover with a thin layer of Na_2CO_3. Place the covered crucible over a moderately low flame, increase the flame gradually to a maximum of 1000-1100°C, and maintain it there until the mass is equiescent (about 45 min.). Transfer the crucible cover to a 300-ml casserole and set aside to cool. Remove the burner, grasp the crucible with the tongs, and slowly rotate it so that the molten contents will spread over the sides and solidify as a thin shell over the interior. Set the crucible aside to cool.

Rinse the outside of the crucible and place it on its side in the 300-ml casserole, adding water to approximately 100 ml, and cover the casserole. Warm the casserole. Very slowly and cautiously add 20 ml of HCl (sp gr 1.19) to the covered casserole. Warm the casserole and stir until the cake disintegrates and can be removed easily. By means of a glass rod, lift the crucible out of the liquid rinsing the outer surface thoroughly with water. Polish and rinse the inner surfaces of the crucible with HCl (1:3); then add the rinse to the casserole. Remove the crucible cover, rinse, and polish as described above. If any gritty particles are present, the fusion is incomplete and must be repeated, using a new sample and a larger quantity of Na_2CO_3.

Complete the determination of SiO_2 in accordance with Section 2 of Method 1101 of Fed. Test Method Std. No. 158, starting with the instruction within paragraph 2.1 of that section: "Evaporate the solution to dryness on a steam bath---". Maintain the rim hot and dry be applying heat from above by means of infra-red lamps to prevent the solution from creeping over the rim of the casserole.

Treat the filtrate reserved from the SiO_2 determination in accordance with Paragraph 3.1.1 and 3.1.3 of Method 1101 of Fed. Test Method Std. No. 158. Reserve the combined filtrate for the determination of MgO. Calculate the combined percentages of SiO_2, Al_2O_3, and Fe_2O_3 by adding the percentage of SiO_2 to the precentage of Al_2O_3 and Fe_2O_3 determined as described above. Report the result to the nearest 0.1 percent.

4.4.2.3.2 Optional method. Weigh out a sample of ignited material equivalent to 0.5 g of moisture-free sample before ignition, calculated as follows:

$$\text{Weight of ignited sample, g} = \frac{0.5\ (100.0 - \text{loss on ignition, percent})}{100}$$

Thoroughly mix the sample with about 0.5 g of NH_4Cl in a 50-ml beaker. Complete the determination of SiO_2 in accordance with Section 2 of Method 1111 of Fed. Test Method Std. No. 158 starting with the instruction within paragraph 2.1 of that section: "Cover the beaker with a watch glass.....". Complete decomposition is not obtained using this method.

Treat the filtrate reserved from the SiO_2 determination in accordance with paragraphs 3.1.1 and 3.1.3 of Method 1101 of Fed. Test Method Std. No. 158. Reserve the combined filtrate for the determination of MgO. Calculate the combined percent of SiO_2, Al_2O_3, and Fe_2O_3 by adding the percentage of SiO_2 to the precentage of Al_2O_3 and Fe_2O_3 determined as described above. Report the result to the nearest 0.1 percent. In the event that the result indicates failure to comply with the specification limit, repeat the determination using the referee method (see 4.4.2.3.1).

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4.4.2.4 Magnesium oxide (MgO).

4.4.2.4.1 Referee method. Determine percentage of MgO in accordance with Section 5 of Method 1101 of Fed. Test Method Std. No. 158 after first removing the ammonium hydroxide group as described in 4.4.2.3.1, and then removing the CaO by the double precipitation procedure described in Section 4 of Method 1101 of Fed. Test. Method Std. No. 158. Also remove the manganese as described in Section 4 of Method 1101 of Fed. Test Method Std. No. 158.

4.4.2.4.2 Optional method. Determine the precentage of MgO in accordance with either Section 5 of Method 1101 or Section 5 of Method 1111 of Fed. Test. Method Std. No. 158 after first removing the ammonium hydroxide group as described in 4.4.2.3.2 and then removing the CaO by the single precipitation procedure described in Section 4 of Method 1111 of Fed. Test Method Std. No. 158. Manganese need not be removed. In the event that the result indicates failure to comply with the specification limit, repeat the determination using the referee failure to comply with the specification limit, repeat the determination using the referee method (see 4.4.2.4.1).

4.4.2.5 Sulfur trioxide (SO_3). Determine the percentage of SO_3 in the pozzolan in accordance with the gravimetric procedure described in Section 2 of Method 1201 of Fed. Test. Method Std. No. 158, except that the pozzolan need not be completely decomposed by the acid.

4.4.2.6 Available alkalies. Thoroughly mix, by grinding together in a mortar or in a mechanical blender, pozzolan and calcium hydroxide ($Ca(OH)_2$) in a proportion of 2-1/2 parts of pozzolan to 1 part of hydrated lime by weight. Add 7 g. of this mixture to 10.1 ml of water in a small plastic vial of approximately 25-ml capacity, stirring until the mixture is uniform. Seal the vial and store at 38°C.

Open the vial at the mixture age of 28 days, and transfer the contents to a 250-ml casserole. Break up and grind the cake with a pestle, adding a small amount of water, if nececcasy, so that a uniform slurry containing no lumps is obtained. Add sufficient water to make a total volume of 200 ml. Let stand 1 hr. at room temperature with frequent stirring. Filter through a medium-testure filter paper into a 500-ml volumetric flask. Wash thoroughly with hot water (eight to ten times). At times it may be necessary to break the vial and peel off the plastic from the solid cake. In such cases, care should be exercised to avoid the loss of material and to remove all solid material from the fragments of the vial. If the cake is too hard to break up and grind in the casserole, a mortar should be used.

Neutralize the filtrate with HCl (1:3) using 1 to 2 drops of phenolphthalein solution as the indicator. Add exactly 5 ml of HCl (1:3) in excess. Cool the solution to room temperature and fill the flask to the mark with distilled water.

Determine the amount of sodium and potassium oxides in the solutions using the procedure described in Method 1401 of Fed. Test. Method Std. No. 158, except that the standard solutions shall be made up to contain 8 ml of $CaCl_2$ stock solution per liter of standard solution, and the solution as prepared shall be used in lieu of the solution of cement. The standard solution made up with 8 ml of $CaCl_2$ stock solution contain the equivalent of 504 ppm CaO. Tests have shown that this amount closely approximates the amount of calcium dissolved in the test solution.

Calculate the results as precentage by weight of the original pozzolan material. Report as equivalent percentage of Na_2O, calculated as follows:

$$\text{Equivalent percent } Na_2O = \text{percent } Na_2O + 0.658 \times K_2O$$

4.4.3 Physical tests (see 3.2).

4.4.3.1 Fineness, surface area. Determine the fineness of the pozzolan in accordance with the procedure described in Method 2101 of Fed. Test. Method Std. No. 158, except that the value for specific gravity, determined in accordance with 4.4.3.8 shall be used in calculating the weight of the sample used for the test. When fineness of the pozzolan is determined, weight of the sample should be such as to give a porosity as near as possible to that of the standard cement sample and yet insure proper compaction. All fineness tests, results of which are to be compared, should be

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made using as nearly as possible the same porosity. Report the porosity, together with the fineness, whenever check tests are to be made on a sample. Use the following equation to calculate the surface area of the pozzolan in square centimenters per cubic centimeter.

$$S_v = \frac{S_s}{\sqrt{t_s}} \cdot \sqrt{t} \cdot \rho_s \cdot \left[\frac{(1-\epsilon_s)\sqrt{\epsilon^3}}{(1-\epsilon)\sqrt{\epsilon_s^3}}\right] \cdot \left[\frac{\sqrt{\eta_s}}{\sqrt{\eta}}\right]$$

where:

S_v = surface area, sq cm/cu cm of test sample.

ϵ = porosity of prepared bed of test sample.

t = measured time interval, sec, of manometer drop for test sample.

S_s = specific surface, sq cm per g, of the standard sample used in calibration of the apparatus.

ρ_s = specific gravity of standard sample used in calibration of apparatus (assumed to be 3.15).

ϵ_s = porosity of prepared bed of standard sample used in calibration of apparatus.

η_s = viscosity of air in poises at the temperature of test of the standard sample used in calibration of the apparatus.

η = viscosity of air in poises at the temperature of test of the test sample.

t_s = measured time interval, sec, of manometer drop for standard sample used in calibration of the apparatus.

When specific surface is determined in square cm per g (S) in accordance with Method 2101 of Fed. Test Method Std. No. 158 the surface area in square centimeters per cubic centimeter can be calculated as follows:

$$S_v = \rho S$$

When the mean particle diameter in microns has been determined, surface area in square centimeters per cubic centimeter can be calculated as follows:

$$S_v = \frac{60{,}000}{d}$$

where:

d = mean particle diameter, microns.

Since the above relations exist, these several measurements of fineness are related as indicated in table IV.

TABLE IV. Mean particle diameter and surface area

Mean Particle Diameter, Microns	Sq cm/cu cm 1/	Surface Area Sq cm/g Specific Gravity			
		2.0	2.5	3.0	3.15
2	30,000	15,000	12,000	10,000	9524
3	20,000	10,000	8,000	6,667	6349
4	15,000	7,500	6,000	5,000	4762
5	12,000	6,000	4,800	4,000	3809
6	10,000	5,000	4,000	3,333	3175
7	8,572	4,286	3,429	2,857	2721
8	7,500	3,750	3,000	2,500	2381
9	6,667	3,333	2,667	2,222	2117

1/ Values in this column also apply as sq cm/g for materials of specific gravity 1.000.

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4.4.3.2 Pozzolanic strength.

4.4.3.2.1 Test specimens. Mold specimens from a control mixture and from a test mixture in accordance with Method 2301 of Fed. Test Method Std. No. 158. The same portland cement shall be used in both the control mixture and the test mixture, and it shall meet the requirements for type II cement of SS-C-192 and have an alkali content, calculated as sodium oxide, between 0.40 and 0.80 percent. If a sample of portland cement of the brand and type to be used on the project is available, it shall be used. In the test mixture, replace 35 percent of the absolute volume of the amount of cement used in the control mixture by an equal absolute volume of pozzolan. The proportion of pozzolan to cement specified for this test is not necessarily the proportion that will be used in the concrete.

Mold at least three 2-inch cube specimens representing each mixture. Three-cube batches shall be made in accordance with table V.

TABLE V. Molded cube specimens

Aggregate and Water	Control Mixture	Test Mixture
Portland cement, g.	250	162.5
Pozzolan, g.	None	A 1/
Graded Ottawa sand, g.	687.5	687.5
Mixing water, distilled, ml 2/	X	Y

1/ $A = \frac{87.5 \text{ specific gravity of pozzolan.}}{\text{specific gravity of portland cement.}}$

2/ Sufficient to produce a flow of 105 to 115 percent.

4.4.3.2.2 Storage of test specimens. After molding keep all of the specimens in the molds on the base plates and immediately place them in a moist room at 73.4° ± 3°F. for 20 to 24 hours with their upper surfaces exposed to moist air, but protected from dripping water. Remove all specimens from the molds 20 to 24 hours after molding and place in a close-fitting metal or glass container, sealed airtight, and stored at 100° ± 3°F. for 27 days. Allow the specimens to cool for 4 ± 1/4 hour to 73.4° ± 3°F. before being tested. Any metal container having a capacity of 3 cubes may be used if it can be sealed airtight, preferably by soldering. Wide-mouth Mason jars of 1-quart capacity have been found to be satisfactory provided care is taken to prevent leakage and breakage.

4.4.3.2.3 Procedure. At an age of 28 days test the specimens for compressive strength according to the applicable provisions of Method 2301 of Fed. Test Method Std. No. 158.

4.4.3.2.4 Calculation. Calculate the pozzolanic strength as follows:

$$\text{Pozzolanic strength, percent} = \frac{\text{Average compressive strength of cubes from test mixture, p.s.i.}}{\text{Average compressive strength of cubes from control mixtures, p.s.i.}} \times 100$$

4.4.3.3 Lime-pozzolan mortar strength.

4.4.3.3.1 Apparatus. The apparatus shall conform to the requirments of Method 2001 of Fed. Test Method Std. No. 158, and, in addition, a cover plate shall be provided for sealing the filled cube molded in accordance with the procedures outlined below.

4.4.3.3.2 Materials.

(a) Pozzolan. The pozzolan shall be oven dried at 105° to 110°C. (221° to 230°F.) to constant weight.

(b) Lime. The lime shall be reagent grade calcium hydroxide, and shall be of such fineness that the residue on a No. 325 (44-micron) sieve is not greater than 5 percent by weight, as determined in accordance with Method 2111 of Fed. Test Method Std. No. 158.

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(c) Sand. The sand shall be natural silica Ottawa sand from Ottawa, Illinois, conforming to the requirements of Method 2001 of Fed. Test Method Std. No. 158 for graded sand.
(d) Water. The water shall be distilled.

4.4.3.3.3 Temperatures and humidity. The temperature and humidity requirements of Method 2001 of Fed. Test Method Std. No. 158 shall be complied with for mixing, molding and storing to an age of 24 ± 2 hours. Subsequent storage shall be at a temperature 130 ± 3°F.

4.4.3.3.4 Number of specimens. At least three specimens shall be made and tested for each lot of pozzolan.

4.4.3.3.5 Proportioning and mixing of mortar. The mortar shall contain, by weight, 2 parts of oven-dry pozzolan, 1 part of calcium hydroxide, 9 parts of graded Ottawa sand, and sufficient water to obtain a flow of 110 plus or minus 5. The flow shall be determined in accordance with paragraph 4.3, Method 2301, Fed. Test Method Std. No. 158. The calcium hydroxide, pozzolan, and sand shall be thoroughly mixed by shaking in a closed container for 2 minutes. Mix the mortar in the mechanical mixer described in paragraph 3.10, Method 2001, Fed. Test. Method Std. No. 158, and as follows. Place the dry paddle and the dry bowl in the mixing position in the mixer. Then introduce the materials for a batch into the mixer and mix in the following manner:

(a) Place all of the mixing water in the bowl.
(b) Start the mixer at slow speed (140 ± 5 rpm). Add the mixture of calcium hydroxide, pozzolan, and sand slowly over a period of 1 minute.
(c) Stop the mixer, change to medium speed (285 ± 10 rpm) and mix for 30 seconds.
(d) Stop the mixer and let the mortar stand for 1-1/2 minutes. During the first 15 seconds of this interval, quickly scrape down into the batch any mortar that may have collected on the side of the bowl; then for the remainder of this interval, cover the bowl with the lid.
(e) Finish by mixing for 1 minute at medium speed (285 ± 10 rpm).

4.4.3.3.6 Preparation of molds and molding specimens. These operations shall be conducted in accordance with the applicable provisions of Methods 2001 and 2301 of Fed. Test Method Std. No. 158.

4.4.3.3.7 Storage of test specimens. Immediately upon completion of molding, cover the specimens with a rigid glass or metal plate and store in the moist room or moist closet at 73.4° ± 3°F. to an age of 24 ± 2 hours. Then remove the specimens, in the molds, from the moist room or moist closet and seal the cover plate to the top of the mold; then store the specimens at 130° ± 3°F. with the cover plate down. At seven days age, remove the specimens from the 130° ± 3°F. storage, allow to cool at 73.4° ± 3°F. for 4 ± 1/4 hour, remove from the molds, and test. The procedure for sealing the cover plate to the mold may be similar to that described in Method 2001 of Fed. Test Method Std. No. 158 for preparing watertight molds.

4.4.3.3.8 Procedure. Test the specimens within 1 hour after removal from the molds. During the interval between stripping and testing, protect the specimens from loss of moisture. The remainder of the procedure shall be in accordance with the applicable provisions of Method 2301 of Fed. Test Method Std. No. 158.

4.4.3.3.9 Calculation, retests, and reports. Calculate the pozzolanic strength and make any necessary retests in accordance with the applicable provisions of Method 2301 of Fed. Test Method Std. No. 158. The value to be reported shall be the arithmetic average of two or more specimens.

4.4.3.4 Soundness. Determine soundness in accordance with Method 2211 of Fed. Test Method Std. No. 158, except that the specimens shall be molded from a paste composed of 25 parts by weight of pozzolan and 100 parts by weight of a portland cement meeting the requirements for type II cement of SS-C-192. If a sample of portland cement of the brand and type to be used on the project is available, it shall be used.

4.4.3.5 Water requirement. Determine compliance with the water requirement provision of the specification as follows: Make a test mixture as described in 4.4.3.2.1, except that it shall have a water content equal to 112 percent (for tests of class N pozzolans) or 103 percent (for tests of class F pozzolans) of the amount of water (X) used in the control mixture described in 4.4.3.2.1.

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Determine the flow of this test mixture. If the flow is at least as great as the flow that was determined for the control mixture described in 4.4.3.2.1 containing X ml of mixing water, the pozzolan has complied with the water requirement provision of the specification. Calculate the value to be reported as follows:

Increase in flow equals flow of pozzolan mortar minus flow of control mortar.

4.4.3.6 Reactivity with cement alkalies. Conduct the reduction of mortar expansion test in accordance with applicable portions of ASTM Designation C 227. The cement used shall conform to the requirements of SS-C-192 for type I or II cement and shall contain at least 1.0 percent total alkalies calculated as Na_2O (percent Na_2O + 0.658 percent K_2O). When a cement of the indicated alkali content is not available, any cement meeting the requirements of SS-C-192 for type I or II cement may be used provided the average expansion of the control specimens in the test for reduction of mortar expansion is equal to or greater than 0.100 percent at 14 days. The aggregate as used shall consist of equal portions by weight of the following five sieve fractions: No. 4 to No. 8, No. 8 to No. 16, No. 16 to No. 30, No. 30 to No. 50, and No. 50 to No. 100. Make specimens from two mixtures: (a) Control specimens prepared in accordance with ASTM Designation C 227, except that the mixture shall contain 400 g of portland cement, 900 g of aggregate composed of crushed and graded Pyrex, and 180 ml of water: (b) test specimens prepared in accordance with ASTM Designation C 227, except that the mixture shall comprise 300 g of portland cement used in preparation of the control specimens, 900 g of the crushed and graded Pyrex, a quantity of pozzolan equal in absolute volume to the absolute volume of 100 g of the portland cement used, and 180 + 8/7 (Y - X) ml of water. For definitions of X and Y, see 4.4.3.2.1. The factor (Y X) may be either positive or negative. The flow test need not be performed on either mixture. Mold three specimens from each mixture. Make an initial reading of length at an age of 24 $\pm$ 2 hours, and a final reading at an age of 14 days. Calculate the reduction of mortar expansion by the formula:

$$R_e = \frac{(E_r - E_t) \times 100}{E_r}$$

where:

R_e = reduction of mortar expansion, %,

E_r = average expansion of bars from the control mixture, and

E_t = average expansion of bars from the test mixture

4.4.3.7 Increase in drying shrinkage of mortar bars. Determine the increase in drying shrinkage of mortar bars containing the pozzolan under test in accordance with the applicable provisions of Sections 23 through 25 of ASTM C 311, except that the control mixture and the test mixture shall have the proportions given in 4.4.3.2. Make one batch of mortar from each mixture, and mold three specimens from each batch. Test the three specimens from each batch concurrently. If available, the cement to be used on the project shall be used in this determination. For purposes of these specifications, calculate the increase in drying shrinkage of mortar bars as follows:

Increase in drying shrinkage of mortar bars, percent = $S_t - S_c$, where S_t and S_c are the average percentage drying shrinkage of test specimens and control specimens, respectively, calculated in accordance with Section 25 of ASTM C 311. Report the result to the nearest 0.01. If S_c is larger than S_t, prefix a minus sign to the change of drying shrinkage of mortar bars reported.

4.4.3.8 Specific gravity (alternate methods).

4.4.3.8.1 LeChatelier flask method. Determine specific gravity of the pozzolan in accordance with the procedures described in Method 2601 of Fed. Test Method Std. No. 158, except that about 45 g of pozzolan, that has been dried in accordance with paragraph 4.4.2.1 above. should be used.

4.4.3.8.2 Volumetric flask method.

4.4.3.8.2.1 Apparatus and materials. Volumetric flask having a capacity of 100 ml; water bath maintained at 25 $\pm$ 0.25 C and equipped with a stirring device; glass desiccator constructed with heavy walls to withstand a vacuum of one atmosphere, and with an opening at the side; motor-driven oil vacuum-type pump capable or reducing the absolute pressure of the system to 3 mm Hg; thermometer having a range of from 0 to 60 C and graduated in 0.1 C divisions; balance sensitive to 0.001 g; and kerosene free of water, or naphtha having a gravity not lighter than 62° A.P.I.

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4.4.3.8.2.2 Specific gravity of kerosene and standardization of volumetric flask. Fill the volumetric flask with freshly boiled distilled water at 23 to 24 C, place it in the water bath, and after temperature equilibrium has been attained, remove excess water with a medicine dropper to bring the level exactly to the 100 ml mark. Stopper the flask, dry, and weigh. Empty, clean, dry, and reweigh the flask. Repeat the process above using kerosene instead of water. Calculate the specific gravity of the kerosene as follows:

$$\text{Specific gravity } 25/25C = \frac{A}{B}$$

where:

A = Weight of kerosene, and
B = Weight of water.

4.4.3.8.2.3 Procedure. Transfer approximately 20 grams of pozzolan to a clean, dry, weighed volumetric flask. Reweigh flask and pozzolan to obtain weight of pozzolan. Add enough kerosene to approximately half fill the flask. Shake gently to insure wetting of all pozzolan. Place the flask in the desiccator. Connect the desiccator to the vacuum pump. Start the pump and continue the evacuation for 1/2 to 1 hour or until no air bubbles rise from the pozzolan. Turn off the pump and slowly admit air into the flask. Remove the flask from the desiccator, fill with kerosene at 24 to 25 C and permit to come to constant temperature at 25 $\pm$ 0.5 C. in water bath. Remove from water bath and immediately remove excess kerosene with a medicine dropper to bring level of kerosene exactly to the 100 ml mark. Stopper the flask, dry and allow to come to room temperature and weigh.

4.4.3.8.2.4 Calculations. Calculate the specific gravity of the pozzolan as follows:

$$\text{Specific gravity of pozzolan} = \frac{P \times S}{(A + C) - F}$$

where:

A = Weight of kerosene to fill flask,
C = Weight of flask and pozzolan,
P = Weight of pozzolan used,
F = Final weight of the flask, with pozzolan and kerosene, and
S = Specific gravity of the kerosene.

4.4.3.8.2.5 Precautions. Before a desiccator is used for the first time, it should be wrapped in a towel and tested under absolute pressure of under 3 mm. Care should be exercised in handling the desiccator when under vacuum, since a sudden jar may cause it to collapse.

4.5 Rejection. The pozzolan may be rejected if it fails to meet any of the following criteria.

4.5.1 Results of consecutive tests. The average of the results of any five consecutive tests for any given requirement shall meet the specified limit. For those tests based on composite samples representing 1000 tons, the average of the results of all tests, when the number of tests from a given source is less than five, shall meet the specified limit.

4.5.2 False set. The pozzolan may be rejected if it is found to produce false set in the concrete in which it is used. When abnormal stiffening characteristics are experienced in either laboratory or field concrete in which the pozzolan is used, tests of the portland cement will be made in accordance with the provisions of the procedure referenced in SS-C-192 for false set. If the portland cement is not shown to be false setting when so tested, the pozzolan will be regarded as causing the false set, unless the abnormal stiffening can be shown to have been caused by the effects of other admixtures such as air-entraining admixtures, water-reducing admixtures, retarding admixtures, or accelerating admixtures.

4.5.3 Weight of packages. Packages varying more than 5 percent from the stated weight may be rejected, and if the average weight of the packages in any shipment, as shown by weighing 50 packages taken at random, is less than that specified, the entire shipment may be rejected.

4.5.4 Pozzolan in storage. Pozzolan remaining in storage prior to shipment for a period longer than 6 months after test may be retested and shall be rejected if more than 1.0 percent is retained on a No. 20 sieve (see 2.1 of Section 5 of Fed. Test Method Std. No. 158) or if it fails to meet the fineness requirement.

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5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. Pozzolan in quantities of less than 100 lb net shall be packed in bags conforming to UU-S-48.

5.1.2 Level B. Pozzolan in quantities of less than 100 lb net shall be packed in bags conforming to UU-S-48.

5.1.3 Level C. Pozzolan shall be pakced in a manner which will insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.2 Marking.

5.2.1 Civil agencies. In addition to markings required by contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. In addition to marking required by contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Pozzolan is intended for use in concrete construction.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Class (see 1.2).
(c) Size of lot, if different from 4.2.1.
(d) Selection of applicable level of packing required (see 5.1).

6.3 Definitions.

6.3.1 Pozzolan. Pozzolan is a siliceous or siliceous and aluminous material which in itself possesses little or no cementitious value, but which will, in finely divided form and in the presence moisture, chemically react with calcium hydroxide at ordinary temperatures to form compounds possessing cementitious properties.

6.3.2 Class N. This class includes most natural pozzolans, such as: some diatomaceous earths; opaline cherts and shales; tuffs and volcanic ashes or pumicites, any of which may or may not be processed by calcination; and various materials requiring calcination to induce satisfactory properties, such as some clays and shales. To be considered as an acceptable class N pozzolan under the provisions of this specification, a material must comply with the applicable requirements for this class as given herein. In general, those materials requiring calcination must be calcined at temperatures in the range of 1400° to 1800°F.

6.3.3 Class F. This class includes fly ash and other materials that comply with the applicable requirements for this class as given herein. Fly ash is the finely divided residue that results from the combustion of ground or powdered coal, and is transported from the combustion chamber by exhaust gases, and which may have been processed subsequently.

Review interest:

Navy - YD

Preparing activity:

Com - NBS

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Order for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price cents each.

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